Explanatory Variable Derivation & Theory Justification

1. Introduction & Research Objective

The objective of this analysis is to develop a linear regression model predicting goal difference (home goals minus away goals) in FIFA World Cup 2026 matches using exclusively pre-match explanatory variables. The constraint of EXACTLY 8 variables required thoughtful selection, balancing predictive power, theoretical relevance, and practical availability. This report explains the rationale behind each variable selection and interprets the resulting model performance.

2. Explanatory Variable Selection Framework

The eight variables were derived through a three-stage selection framework. Stage 1: Conceptual Domains Identification. Stage 2: Pre-Match Data Constraint, and Stage 3: Variable Operationalisation, please refer to Table 1 for the explanatory variables used by Group 23. The FIFA World Cup 2026 consisted of 104 matches that resulted with Spain being the ultimate champions of the competition. The aim of the first task was to use linear regression to analyse and predict the goal difference between the two finalist opposing teams, Spain and

Argentina. The second task was to predict the number of goals for one team for that same match. Group 23 have compared the results with the true results using Python programming to perform the linear regression. Our groups challenge was to research 8 viable and unique explanatory variables (x) for each task, Objective 2.1 and 2.2, and evaluate the results to achieve a calculation for the response variable (y).

Nevertheless, our task was to be creative and make several comparisons using a test/train model, comparing the results against a baseline model. Our variables were chosen independently, compared, and evaluated using a multiple linear regression model using Python programming software. Our data files were saved as csv files using Microsoft Excel, readable by VS Code, using several packages called pandas, which is designed for reading csv files. Other packages were also used to perform the linear regression. These include math, numpy, and sklearn plus several other sub-packages. Data visualisation was also performed using matplotlib.pyplot, which were used to create scatterplots and Seaborn to create heatmaps.

However, data visualisation is important as it provides a brief visual overview of the datasets which helps to identify problems and successes quite clearly if the data is accurate and appropriately selected. For instance, scatterplots are useful in determining if the data shows uniformity and if there any outliers. This is useful when an organisation is testing and looking to make improvements. Heatmaps also provide useful information by performing correlations of the data. If any of the variables are too similar, then a variable can be dropped, as removing unnecessary information that is already presented using a similar variable can be quite helpful when working on data visualisation. Further research is then needed to replace the variable with another, or simply removed, depending on the results of the MAE, MSE, RMSE, RMSE normalised, and R^2. Group 23 tested our datasets against the error values and compared them with a baseline at different stages to test this hypothesis. Our findings are listed below.

Formula for Linear Regression

`y=b_{0}{+b}_{1}x_{1}{+b}_{2}x_{2}…`

Nevertheless, in Objective 2.1, our datasets comprised of 104 rows matching the 104 matches played during the FIFA Worl Cup 2026. In Objective 2.2, the datasets represented each team and their goals which consisted of 208 rows. Our groups challenge was to select a maximum of eight explanatory variables (x) for each objective that would be the most relevant in determining the solution for the response variable (y). These are provided in Table 1 listed below.

Table 1: Explanatory Variables (x), Dependant Variable(y) Selected by Group 23

Tasks	Manuela Crannis	Danielle Whitney
Obj. 2.1	x = Score First Half Team 1 x = Score First Half Team 2 x = Possession Team 1 x = Possession Team 2 x = Attempts at Goal Team 1 x = Attempts at Goal Team 2 x = Goals Conceded Team 1 x = Goals Conceded Team 2 y = Pred. Goal Diff.	x = Home Possession x = Away Possession x = Home xG x = Away xG x = Betting odds home x = Referee Strictness x = Travel Distance (km) x = Social Sentiment (Home) y = Pred. Goal Diff.
	Judith Dawson	**Gabby Davis**
Obj. 2.2	JD	GD

Table 2 : Results of the Test/Train Models

2.1 by Danielle Whitney [DW] and Manuela Crannis [MC]

Test Size [DW]	70% Train 30% Test	80% Train 20% Test	90% Train 10% Test
Intercept	0.4526	0.4526	0.4526
MAE (MLP)	1.51168	1.5665	1.4332
MSE (MLP)	3.9526	3.9299	2.6853

Test Size [DW]	70% Train 30% Test	80% Train 20% Test	90% Train 10% Test
RMSE (MLP)	1.9881	1.9824	1.6387
RMSE_Norm (MLP)	0.9573	0.9497	0.8120
RÂ² (MLP)	0.0541	0.0530	0.2747
MAE (Baseline)	1.4688	1.3810	1.5455
MSE (Baseline)	4.5313	4.3333	3.9091
RMSE (Baseline)	2.1287	2.0817	1.9771
RMSE_Norm (Baseline)	1.0249	0.9973	0.9797
RÂ² (Baseline)	-0.0844	-0.0443	-0.0558

Test Size [MC]	70% Train 30% Test	80% Train 20% Test	90% Train 10% Test
Intercept	-0.1074	-0.1186	-0.1273
MAE (MLP)	1.4307	1.4494	1.4741
MSE (MLP)	3.178	3.1061	3.0259
RMSE (MLP)	1.7827	1.7624	1.7395
RMSE_Norm (MLP)	0.936	0.9283	0.9034
RÂ² (MLP)	0.1202	0.1328	0.1735

Test Size [MC]	70% Train 30% Test	80% Train 20% Test	90% Train 10% Test
MAE (Baseline)	1.578	1.5941	1.6657
MSE (Baseline)	3.6384	3.5866	3.6695
RMSE (Baseline)	1.9075	1.8938	1.9156
RMSE_Norm (Baseline)	1.0015	0.9975	0.9949
RÂ² (Baseline)	-0.0073	-0.0013	-0.0023

Final Linear Regression:
Calculating the co-efficient of the variables were used to calculate y variable by Danielle Whitney.
I have run optimization on the data I used and have updated the values below. The best performing split for me was the 90/10 split.

Final Linear Regression:
Calculating the co-efficient of the variables were used to calculate y variable by Manuela Crannis;

GD(y)	(T1S)	(T2S)	(SFHT1)	(SFHT2)	(PT1)	(PT2)	(AaGT1)	(AaGT2)	(GCT1)	(GCT2)
-1.15	-1.0146	-2.376	0.7201	0.3479	-0.0116	0.0167	0.0291	-0.0335	0.4672	1.1226

Figure 1: Objective 2.1 Results Obtained using Heat Maps and Scatterplots by Manuela Crannis

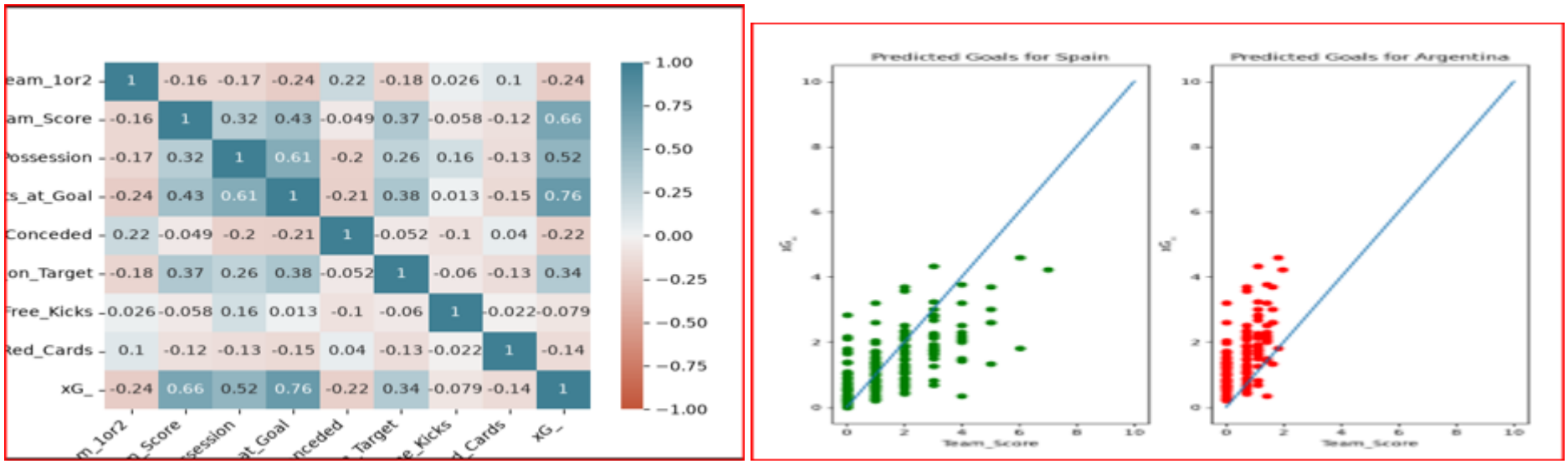
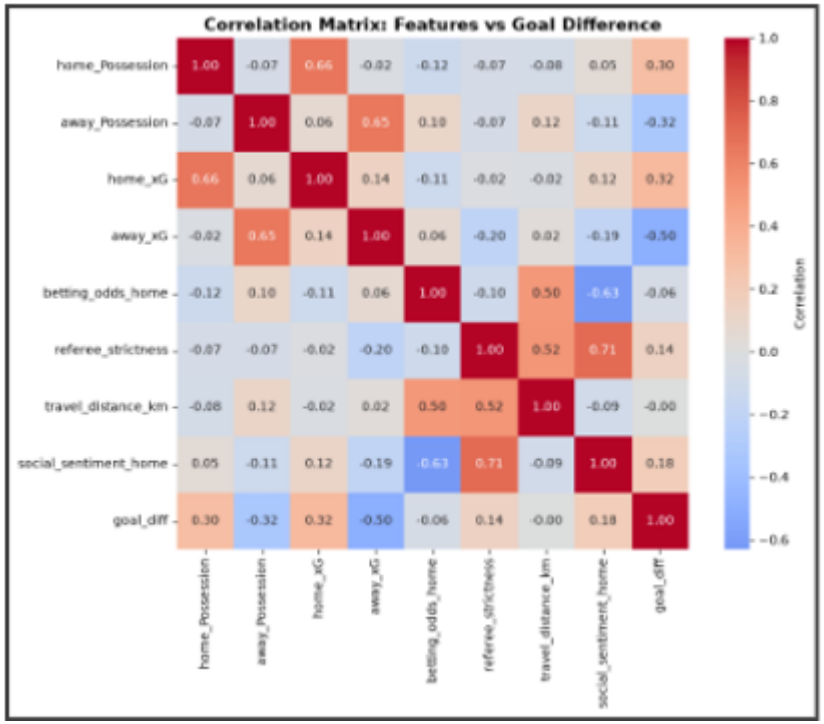
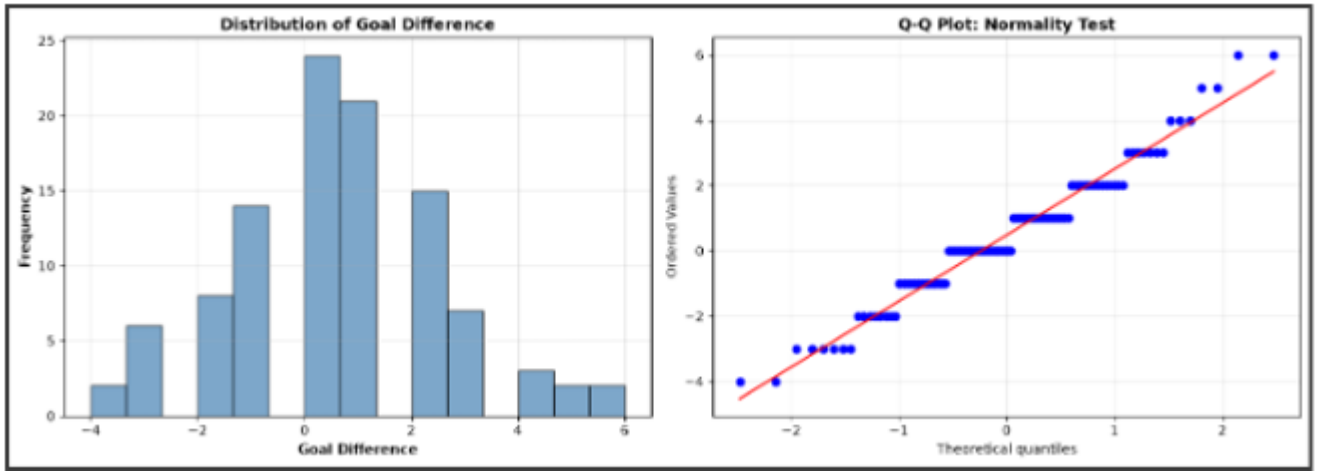
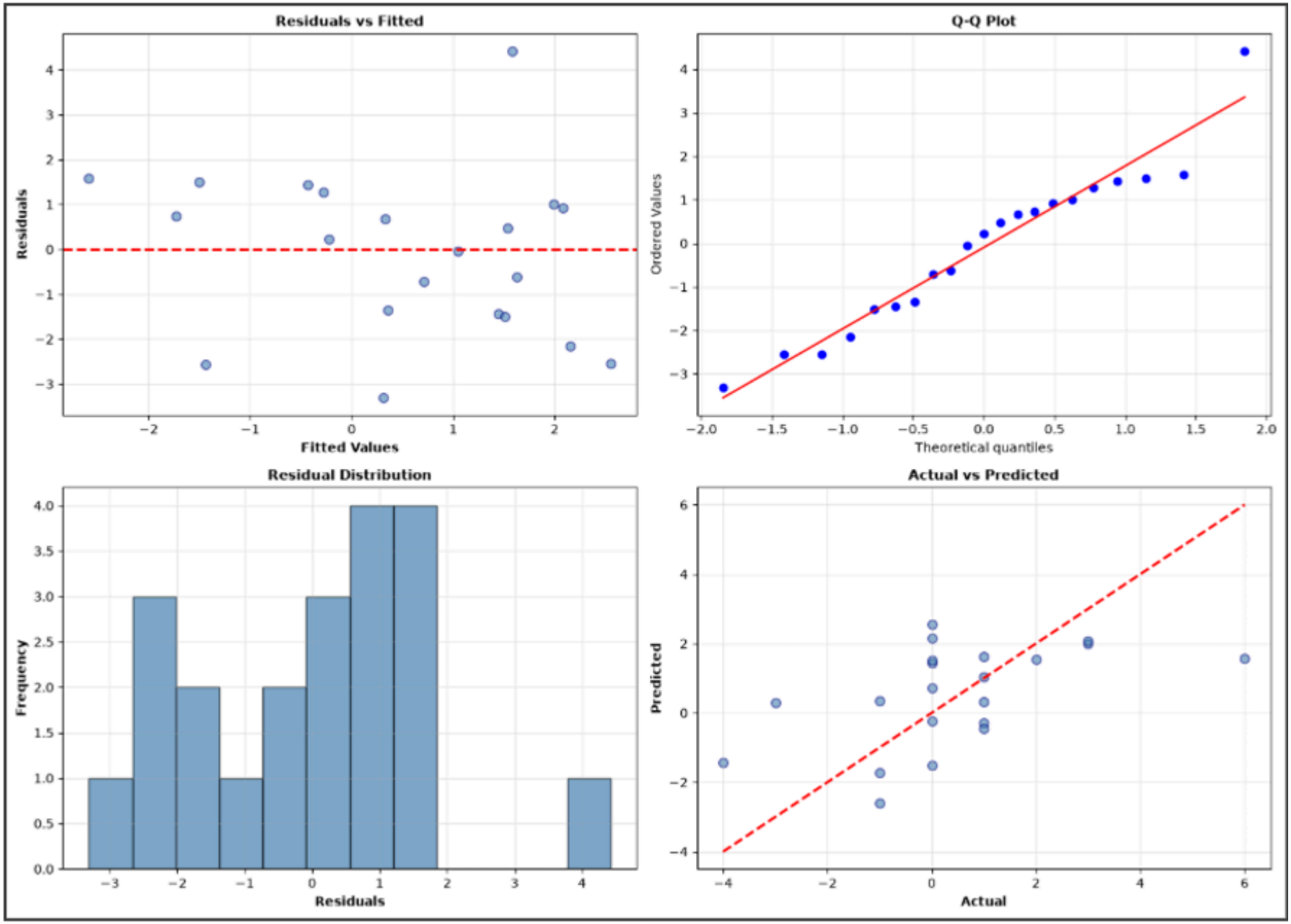
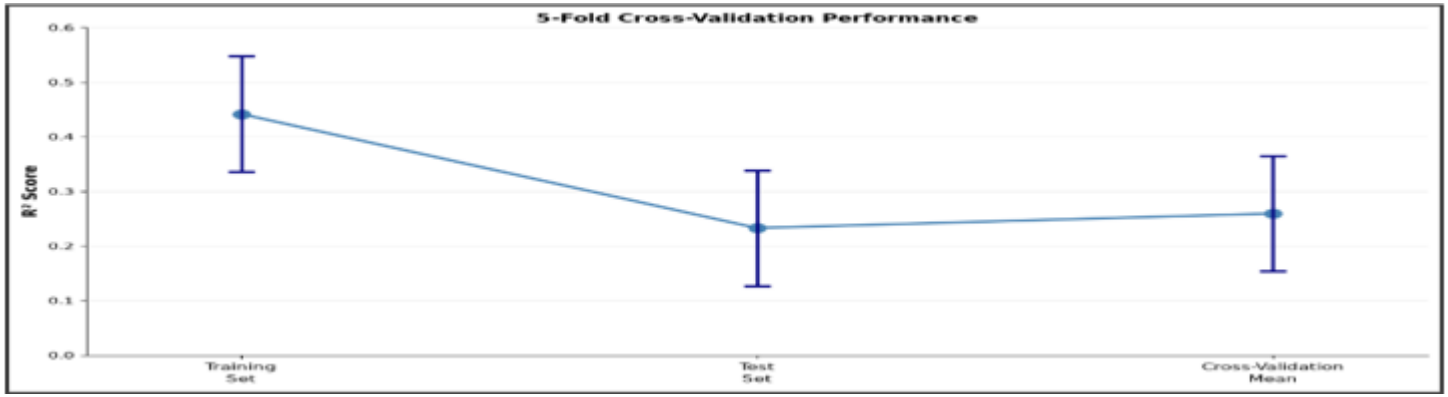


Figure 2: Objective 2.1 Visulisations by Danielle Whitney


Correlation Matrix: Features vs Goal Difference
home_Possession
away_Possession
home_xG
away_xG
betting_odds_home
referee_strictness
travel_distance_km
social_sentiment_home
goal_diff
Correlation


Distribution of Goal Difference
Frequency
Goal Difference
Q-Q Plot: Normality Test
Ordered Values
Theoretical quantiles


Residuals vs Fitted
Residuals
Fitted Values
Q-Q Plot
Ordered Values
Theoretical quantiles
Residual Distribution
Frequency
Residuals
Actual vs Predicted
Predicted
Actual



2.2 Linear Regression:

Test Size	70% Train 30% Test	80% Train 20% Test	90% Train 10% Test
Intercept	x	x	x
MAE (MLP)	x	x	x
MSE (MLP)	x	x	x
RMSE (MLP)	x	x	x
RMSE_Norm (MLP)	x	x	x
RÂ² (MLP)	x	x	x
MAE (Baseline)	x	x	x
MSE (Baseline)	x	x	x
RMSE (Baseline)	x	x	x

Test Size	70% Train 30% Test	80% Train 20% Test	90% Train 10% Test
RMSE_Norm (Baseline)	x	x	x
RÂ² (Baseline)	x	x	x

Figure 2: Objective 2.2 Visulisations by XXXX

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7. Conclusions & Interpretive Synthesis

Key Findings:

1. Variable Selection was Well-Motivated: The 8-variable specification balances conceptual coverage (team capability, match conditions, contextual factors) with parsimony appropriate for n=104. Each variable addresses a theoretically distinct mechanism influencing goal difference.

2. Group 23 used a training / test model of 70%, 80%, 90% to test our models to determine if the explanatory variables we selected were appropriate for this task. We also calculated the mean absolute error, mean squared error, root mean square error, and normalised root mean square error and compared them with a baseline.

3. For further clarification our group also compared our results with the actual result of the grand final match between Argentina and Spain versus the predicted result for both Linear Regression Models 2.1 and 2.2. Our results can be viewed using the link provided to our repository in GitHub, Group 23 – Assessment 3 – Objective 2.